Chapter 15: Incomplete Information, Dynamic Games

ECON 5001
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Based on: *Game Theory* by Giacomo Bonanno

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Where We Left Off

Chapter 14: static games with incomplete information

- Harsanyi transformation $\rightarrow$ a game with Nature
- Solution: BNE $=$ NE of that game

Chapter 15: **dynamic** games with incomplete information

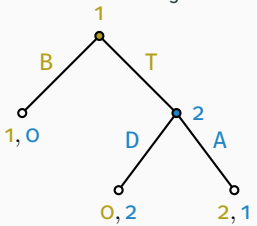
- Same Harsanyi transformation
- But NE allows silly choices off the path (Chapter 4!)
- So use **WSE** (or SPNE) of the Harsanyi game

Big applications: **reputation** and **signaling**

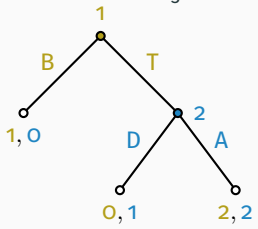
Example: Unsure of P2's Payoffs

P1 knows her own payoffs. P1 thinks P2's payoffs are one of these:

α (prob. $\frac{1}{3}$)



β (prob. $\frac{2}{3}$)

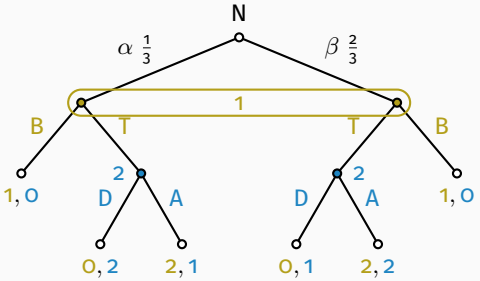


P2 knows which one. True state: α

Backward induction if P1 knew: at α play ____. At β play ____

Bonanno Figures 15.1–15.3.

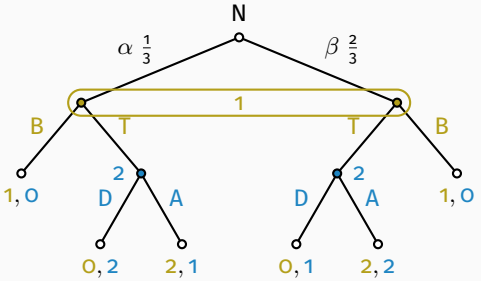
The Harsanyi Game



1 \ 2	DD	DA	AD	AA
B	1, 0	1, 0	1, 0	1, 0
T	0, $\frac{4}{3}$	$\frac{4}{3}$, 2	$\frac{2}{3}$, 1	2, $\frac{5}{3}$

P2's strategy: (at α, at β)

The Harsanyi Game

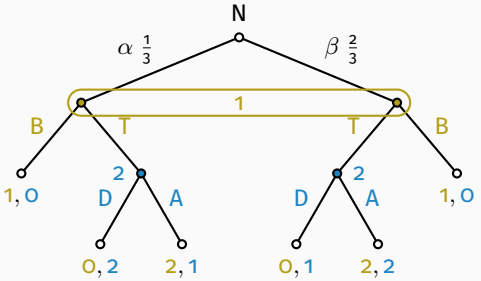


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BNE: _____ _____ _____

The Harsanyi Game



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P2's strategy: (at α, at β)

BNE: _____ _____ _____

Which ones have P2 doing something silly at a node? _____

Use WSE Instead

In dynamic games: BNE can rely on **incredible threats**

- P2 is at a **singleton** node. P2 knows the state!
- (B, DD): at β, P2 "plans" D: 1 instead of 2. Silly
- (B, AD): silly at **both** nodes

From now on: WSE (here it's the same as SPNE) $\Rightarrow$ _____

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True state α: what actually gets played? _____
Backward induction in the true game says ____

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Lesson: P1's uncertainty can lead to play that looks "wrong" in the true game

Reputation

MobLab: “Centipede Game”, Again!

- MobLab time!
- Same game as Chapter 3: Invest or Cash Out
- You now know what backward induction says
- Play it the way you’d actually play it

Write down: what did you expect your partner to do?

Feel free to talk!

Recall: The Chain Store (Chapter 3)

IKEA at Polaris. Wayfair decides whether to open across I-71

- Out: Wayfair 1, IKEA 5. Fight: (0, 0). Share: (2, 2)

Then the same thing in Cincinnati, with another entrant

IKEA's argument: "Fight in Columbus. Cincinnati is watching!"

- Fight, scare off Cincy: $0 + 5 = 5$ Share, Cincy enters: $2 + 2 = 4$

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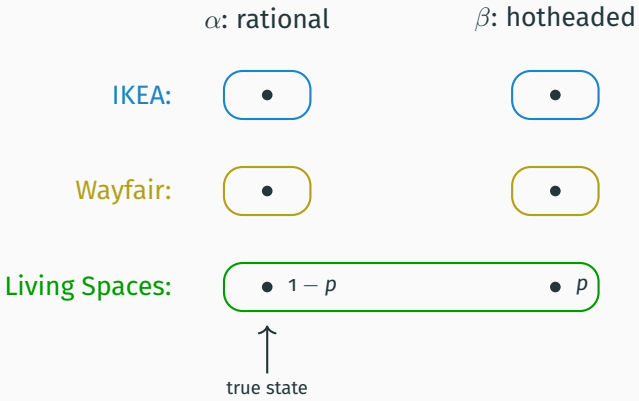
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Deck 05: "The reputation story needs **uncertainty**… Chapter 15"

Add a Little Doubt

Two possible types of IKEA:

- **Rational**: fight gives 0, share gives 2 (as before)
- **Hotheaded**: **likes** fighting. Fight gives 3, share gives 2



- IKEA really is rational
- Wayfair knows it
- Living Spaces (Cincy) isn't sure: hotheaded with prob. p

Bonanno §15.1 (Figures 15.6–15.11) uses slightly different numbers.

Solving It: Start in Cincinnati

If Living Spaces enters Cincinnati:

- Rational IKEA: _____ Hotheaded IKEA: _____

Living Spaces sees what happened in Columbus. Three possible histories:

In Columbus...	Pr(hotheaded)	Living Spaces
Wayfair stayed Out		
Wayfair In, IKEA Shared		
Wayfair In, IKEA Fought		

Candidate: IKEA (both types) would fight in Columbus. Wayfair stays out

Checking the WSE

Living Spaces after Out: belief $=$ prior p (Bayes)

$$\underbrace{p(0) + (1-p)(2)}_{\text{enter}} \leq \underbrace{1}_{\text{stay out}} \iff p \geq _____$$

After Share / Fight: off the path. WSE allows any beliefs. These are sensible

IKEA, if Wayfair entered Columbus:

- Rational: fight _____ vs. share _____
- Hotheaded: fight _____ vs. share _____

Wayfair: in ____ vs. out ____

Reputation Works!

If Living Spaces thinks IKEA might be hotheaded ($p \geq \frac{1}{2}$):

- IKEA's threat to fight in Columbus is **credible**
- Even though Wayfair and IKEA both **know** IKEA is rational!
- IKEA fights to protect a reputation it doesn't deserve

Not in the book: Kreps & Wilson (1982); Milgrom & Roberts (1982); Kreps, Milgrom, Roberts & Wilson (1982).

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Compare Repeated Games: finite PD unravels with **complete** info

- A small chance of a "crazy" type can stop the unraveling
- Same idea explains passing in the centipede game

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It's also a **sequential equilibrium** (Bonanno Exercise 15.2)

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Back to the Centipede

	Chapter 3 (before BI)	Today (after BI)
Avg. turns before Cash Out		
% Cash Out right away		

If a few people **like** to invest, even selfish players should invest for a while

Not in the book. McKelvey & Palfrey (1992, *Econometrica*).

This Is Everywhere

Building a reputation for toughness:

- **Predatory pricing**: fight early entrants to scare later ones
- **Hostages**: “we never negotiate with kidnappers”
- **Bosses**: fire the first slacker to keep the rest in line
- **Central banks**: fight inflation hard to be believed next time
- **Parents**: follow through on the threat… this time

Real versions: many periods, many types, people learning slowly

- Same core: act tough now so others’ **beliefs** about you change

Signaling

Why Do Strikes Happen?

A firm and a union negotiate wages. Then they strike. Then they agree

- Strikes are costly for **both**: lost wages, lost production
- Why not just agree on day one?

Bonanno §15.1, Figures 15.12–15.15.

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Bonanno's answer: **incomplete information**

- The firm knows its profits. The union doesn't
- Enduring a strike is a **credible signal** that profits are low

Bonanno §15.1, Figures 15.12–15.15.

The Setup

- Union asks for a high wage $w_H = 60$ or low wage $w_L = 50$
- Firm says Yes (contract signed) or No
- No $\Rightarrow$ strike for one period. Then the union asks again
- No again $\Rightarrow$ no deal. Both get 0
- Waiting is costly: $\delta = 0.6$

Firm's profit (before wages): **High** $\pi_H = 100$ (prob. 0.7) or **Low** $\pi_L = 55$

- Firm knows. Union doesn't. Actually: Low
- Low firm can't afford the high wage: $55 - 60 < 0$

Last period is easy: firm accepts any wage it can afford

A Separating WSE

Candidate: Union asks 60. High firm says Yes. Low firm says No (strike!) Then union asks 50. Low firm says Yes

Union after a strike: by Bayes, Pr(Low) = _____ $\Rightarrow$ ask 50 ✓

High firm won't fake being Low:

$$\underbrace{100 - 60}_{\text{pay 60 now}} \geq \underbrace{0.6\,(100 - 50)}_{\text{strike, then pay 50}} \qquad _____ \geq _____ \checkmark$$

Low firm: Yes to 60 gives ____. No gives $0.6(55 - 50) =$ ____

Union, first offer: ask 50 $\rightarrow$ 50. Ask 60 $\rightarrow$ $0.7(60) + 0.3(0.6)(50) =$ _____

What Happened?

The true firm is Low $\Rightarrow$ **a strike actually happens**

- Inefficient: both lose from the delay
- But it's how the Low firm **proves** it's Low
- A High firm wouldn't sit through a strike just to save 10

Separating: types choose differently, so actions reveal types

Signal: works only if it's **too costly to fake** for the other type

Signals Everywhere

Costly actions that are cheaper for the “good” type:

- **Education**: a degree is easier for productive workers (Spence)
- **Warranties**: cheap to offer if your product rarely breaks
- **Ads**: only worth paying for if customers come back
- **Engagement rings**: expensive on purpose
- **Peacocks**: a big tail is only survivable if you’re strong

Real versions: continuous signals, many types, noisy signals

- Same core: the cost has to differ across types

Not in the book. Spence (1973, *QJE*). 2001 Nobel Prize.

Truth in Advertising

Gas station sign: “Octane: at least 85.” The truth is 89. True, but vague

- Law: no false claims. But vague claims are legal
- Seller knows quality. Buyer doesn’t

Skeptical buyer: assume the **worst** quality consistent with the claim

- “At least 85” $\Rightarrow$ “it’s 85”
- So a seller with 89 wants to say “89”
- And then 88 does too, and 87...

WSE: buyer ends up as if fully informed. No news is bad news!

Bonanno §15.1 (Milgrom & Roberts, 1986). See Exercise 15.4 for a naive buyer.

Let’s Work One Out

The Honors Section

A senior is a **Star** (prob. $\frac{1}{3}$) or **Average**. They know which

- First: take a hard **Honors** capstone, or **Not**
- Honors costs a Star 1, an Average student 4 (more stress, worse grade)

Nationwide sees the transcript (Honors or not), not the type

- Hires as **Manager** or **Analyst**
- Student gets 5 as Manager, 2 as Analyst, minus Honors cost
- Nationwide: Manager pays off 4 with a Star, 0 with Average. Analyst: 2 either way

Honors doesn't make anyone better at the job. It's only a signal.

The Honors Section

1. Draw the Harsanyi tree. How many info sets does Nationwide have?
2. For which beliefs does Nationwide hire a Manager?
3. Find a **separating** WSE. Check every type and info set
4. Find a **pooling** WSE where nobody takes Honors. What beliefs after Honors?
5. Are those beliefs reasonable? Would an Average student ever take Honors?
6. If Honors cost an Average student only 2, which WSE survive?

What Have We Learned, Charlie Brown?

- Dynamic + incomplete info: Harsanyi game, then **WSE**
  - BNE alone allows incredible threats
- **Reputation**: a little doubt about your type makes threats credible
  - Fixes the chain-store paradox
- **Signaling**: costly actions reveal types, if too costly to fake
  - Strikes, education, warranties
- Skeptical buyers + no lying $\Rightarrow$ full disclosure

Reading & Practice

- Bonanno, Chapter 15
  - §15.1: read
  - §15.2 (bargaining, two-sided): skim
- Exercises: §15.3.1 (15.1, 15.3a, 15.5), §15.3.2 (15.6)
- 15.2 and 15.3b need sequential equilibrium (Ch. 12). 15.7 is a challenge
- Solutions in §15.4. Try them first